

PART : MATHEMATICS

- 1 If $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and B be a 3×3 matrix whose entries are $\{1, 2, 3, 4, 5\}$, then the number of possible

matrices B such that $AB = BA$ are :

- (1) 240 (2) 320 (3) 120 (4) 100

Ans. (3)

Sol. $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Let $B = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$

Now $AB = BA$

$$\Rightarrow \begin{bmatrix} p & q & r \\ a & b & c \\ x & y & z \end{bmatrix} = \begin{bmatrix} b & a & c \\ q & p & r \\ y & x & z \end{bmatrix}$$

$$\Rightarrow p = b, a = q, r = c, x = y, \& z = z$$

Hence number of such matrices are $5! = 120$

- 2 The number of all possible numbers less than 10000 that can be formed using the digits 0, 2, 4, 6, 8 (without repetition) are :

Ans. 165

Sol. 1 digit numbers = 5

$$2 \text{ digit numbers} = \underline{4} \underline{4} = 16$$

$$3 \text{ digit numbers} = \underline{4} \underline{4} \underline{3} = 48$$

$$4 \text{ digit numbers} = \underline{4} \underline{4} \underline{3} \underline{2} = 96$$

$$\text{Total} = 5 + 16 + 48 + 96 = 165$$

- 3 If a circle $36x^2 + 36y^2 - 108x + 120y + c = 0$ neither touches nor cuts the co-ordinate axis and point of intersection of lines $x - 2y = 4$ and $2x - y = 5$ also lies inside the circle, then the value of c lies in :

- (1) $81 < c < 156$ (2) $100 < c < 156$ (3) $81 < c < 150$ (4) $100 < c < 150$

Ans. (2)

Sol. Intersection point of $2x - y = 5$ and $x - 2y = 4$ is $(2, -1)$

So, $(2, -1)$ lies inside the circle $\Rightarrow S_1 < 0$

$$36(2)^2 + 36(-1)^2 - 108(2) + 120(-1) + c < 0$$

$$c < 156 \dots\dots(i)$$

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∴ circle $36x^2 + 36y^2 - 108x + 120y + c = 0$ neither touches nor cuts the co-ordinate axis so

$$g^2 - c < 0 \Rightarrow \left(\frac{-3}{2}\right)^2 - \frac{c}{36} < 0 \Rightarrow c > 81 \dots\dots\dots (ii)$$

$$\text{and } f^2 - c < 0 \Rightarrow \left(\frac{5}{3}\right)^2 - \frac{c}{36} < 0 \Rightarrow c > 100 \dots\dots\dots (iii)$$

From (i), (ii) and (iii)

$$100 < c < 156$$

4. If line $2x + y = k$, ($k < 0$) is a tangent to both the curves $x^2 - y^2 = 3$ and $y^2 = \alpha x$, then the value of α is

Ans. 24.00

Sol. Given slope of line (m) = -2

slope form of tangent to the curve $x^2 - y^2 = 3$ is $y = mx \pm \sqrt{a^2m^2 - b^2}$

$$\Rightarrow y = -2x \pm 3$$

On comparing, with the equation $2x + y = k$, ($k < 0$) $\Rightarrow k = -3$

Now, slope form of tangent to the parabola $y^2 = \alpha x$ is $y = mx + \frac{\alpha}{4m}$

But $m = -2$ so

$$y = -2x + \frac{\alpha}{4(-2)} \Rightarrow -3 = \frac{\alpha}{4 \times (-2)}$$

$$\alpha = 24$$

5. The number of all possible values of $n \in \{1, 2, 3, \dots, 100\}$ which satisfy the condition $11^n > 10^n + 9^n$ is:

Ans. (96)

Sol. Let $11^n > 10^n + 9^n$ $n \in \{1, 2, 3, \dots, 100\}$

$$\Rightarrow 11^n - 9^n > 10^n$$

$$\Rightarrow (10+1)^n - (10-1)^n > 10^n$$

$$\Rightarrow 2 [{}^nC_1 10^{n-1} + {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots] > 10^n$$

$$\Rightarrow \frac{1}{5} [{}^nC_1 10^n + {}^nC_3 10^{n-2} + {}^nC_5 10^{n-4} + \dots] > 10^n$$

$$\Rightarrow \frac{1}{5} [{}^nC_1 + {}^nC_3 10^{-2} + {}^nC_5 10^{-4} + \dots] > 1$$

Clearly the above inequality is true for $n \geq 5$

$$\text{For } n = 4 \text{ we have } \frac{1}{5} \left[4 + \frac{4}{10^2} \right] = \frac{4}{5} \left(\frac{101}{100} \right) < 1, \quad \text{Rejected}$$

Hence, number of such $n \in \{1, 2, 3, \dots, 100\}$ is equal to 96

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6. If an AP, $S_{10} = 530$ & $S_5 = 140$ (where S_n denotes the sum of first n terms of an AP), then the value of

$S_{20} - S_6$ is

- (1) 1562 (2) 1862 (3) 1762 (4) 1662

Ans. (2)

Sol. $S_{10} = 530$

$$\frac{10}{2}[2a + 9d] = 530$$

$$2a + 9d = 106 \quad \dots (1)$$

$$S_5 = 140$$

$$\frac{5}{2}[2a + 4d] = 140$$

$$2a + 4d = 56 \quad \dots (2)$$

$$5d = 50$$

$$d = 10$$

$$a = 8$$

Now,

$$S_{20} - S_6 =$$

$$10[2a + 19d] - 3[2a + 5d]$$

$$14a + 175d$$

$$14 \times 8 + (175)10 = 1862$$

7. The value of r for which the term independent of x in the expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180 is:

- (1) 7 (2) 8 (3) 9 (4) 10

Ans. (2)

Sol. $T_{k+1} = {}^{10}C_k (2x^r)^{10-k} (x)^{-2k} \Rightarrow {}^{10}C_k 2^{10-k} \cdot x^{10r-rk-2k}$

$$\text{Now, } 10r - rk - 2k = 0 \Rightarrow r = \frac{2k}{10-k}$$

$$\text{And } {}^{10}C_k (2)^{10-k} = 180 \Rightarrow k = 8$$

$$r = \frac{2 \times 8}{10-8} = 8$$

8. If $f(x) = \begin{cases} -\frac{4}{3}x^3 + 2x^2 + 3 & ; x > 0 \\ 3xe^x & ; x \leq 0 \end{cases}$ then interval in which $f(x)$ is increasing, is :

- (1) $\left(-1, \frac{3}{2}\right)$ (2) $(0, 1)$ (3) $\left(\frac{-3}{2}, 1\right)$ (4) $\left(\frac{1}{2}, 2\right)$

Ans. (B)

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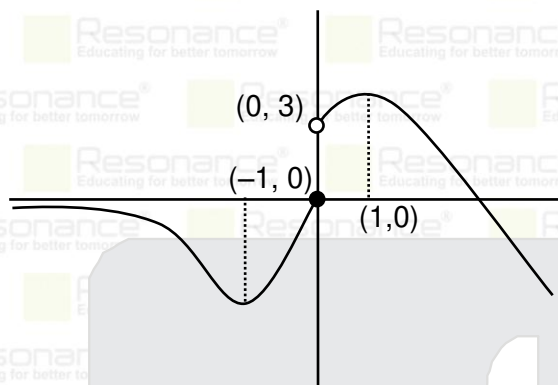
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Sol. $f'(x) = \begin{cases} -4x^2 + 4x, & x > 0 \\ 3(xe^x + e^x), & x \leq 0 \end{cases}$



9. If $z^2 + 3\bar{z} = 0$ has n solutions, then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is :

(1) $\frac{3}{4}$

(2) $\frac{4}{3}$

(3) $\frac{5}{5}$

(4) $\frac{1}{2}$

Ans. (2)

Sol. Let $z = x + iy$

$$(x + iy)^2 + 3(x - iy) = 0$$

$$x^2 - y^2 + 2ixy + 3x - 3iy = 0$$

$$x^2 - y^2 + 3x = 0 \text{ \& } 2xy - 3y = 0$$

Case-1: $y = 0$

$$x^2 - y^2 + 3x = 0$$

$$\Rightarrow x = 0 \text{ or } x = -3$$

Solutions are $z=0$ and $z=-3$

Case-2: $x = \frac{3}{2}$

$$x^2 - y^2 + 3x = 0$$

$$\Rightarrow y = \frac{3\sqrt{3}}{2} \text{ or } y = \frac{-3\sqrt{3}}{2}$$

$$\text{Solutions are } z = \frac{3}{2} + i\frac{3\sqrt{3}}{2} \text{ and } z = \frac{3}{2} - i\frac{3\sqrt{3}}{2}$$

Total number of solutions = $n = 4$

$$\text{So } \sum_{k=0}^{\infty} \frac{1}{4^k} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3} \text{ Ans.}$$

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10 The number of solutions of the equation $\sin^7 x + \cos^7 x = 1$ in the interval $[0, 4\pi]$ is :

- (1) 7 (2) 9 (3) 14 (4) 5

Ans. (4)

Sol. $\sin^2 x + \cos^2 x = 1$, $\sin^2 x \leq 1$ and $\cos^2 x \leq 1$

$$\sin^7 x \leq \sin^2 x$$

$$\cos^7 x \leq \cos^2 x$$

$$\text{so, } \sin^7 x + \cos^7 x \leq 1$$

$$\sin^7 x + \cos^7 x = 1 \text{ when } \sin^7 x = \sin^2 x \text{ \& } \cos^7 x = \cos^2 x$$

$$\text{Case-1 : } \sin x = 0, \cos x = 1 \Rightarrow x = 0, 2\pi, 4\pi$$

$$\text{Case-2 : } \sin x = 1, \cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{5\pi}{2}$$

Total number of solutions = 5

11. The sum of all natural numbers belonging to the set $\{1, 2, 3, \dots, 100\}$, whose HCF with 2304 is 1, is

- (1) 2449 (2) 1633 (3) 1449 (4) 2633

Ans. (2)

Sol. $2304 = 2^8 \cdot 3^2$

Hence n can not be multiple of 2 or 3

Then sum is

$$\Rightarrow n(1) - (n(2) + n(3) - n(6))$$

(where n(a) means the sum of all numbers belonging to the set $\{1, 2, 3, \dots, 100\}$, which are divisible by a)

$$\Rightarrow \frac{100 \times 101}{2} - \frac{2 \times 50 \times 51}{2} - 3 \times \frac{33 \times 34}{2} + 6 \times \frac{17 \times 16}{2}$$

$$\Rightarrow 5050 - 2550 - 1683 + 816$$

$$\Rightarrow 1633$$

12. If $f(x)$ is a continuous function defined as $f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} \ln \left(\frac{1 + \alpha x e^x}{(1 + x e^x)^2} \right) & ; x < 0 \\ \alpha & ; x \geq 0 \end{cases}$, then the value

of α is

- (1) $-\frac{2}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $-\frac{1}{3}$

Ans. (1)

$$\text{Sol.} = \lim_{x \rightarrow 0^-} \frac{x^3}{(1 - \cos 2x)^2} \ln \left(\frac{1 + \alpha x e^x}{(1 + x e^x)^2} \right)$$

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$$= \lim_{x \rightarrow 0^-} \frac{x^3 \times x}{4 \sin^4 x} \frac{\ln(1 + \alpha x e^x) - 2 \ln(1 + x e^x)}{x}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{4} \frac{\ln(1 + \alpha x e^x) - 2 \ln(1 + x e^x)}{x}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{4} \left\{ \frac{\ln(1 + \alpha x e^x) \alpha e^x}{\alpha e^x} - \frac{2 \ln(1 + x e^x) e^x}{\alpha e^x} \right\}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{4} \{ \alpha e^x - 2 e^x \}$$

$$= \frac{1}{4} (\alpha - 2)$$

$$\text{Now, } \lim_{x \rightarrow 0^+} \alpha = \alpha$$

$\therefore f(x)$ is continuous function so

$$\frac{\alpha - 2}{4} = \alpha \Rightarrow \alpha = \frac{-2}{3}$$

13. If the domain of $f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x - 1}{2} \right)}}$ is $(\alpha + \beta]$, then the value of $\alpha + \beta$ is :

(1) $\frac{1}{2}$

(2) $\frac{3}{2}$

(3) 1

(4) 2

Ans. (2)

Sol. $0 \leq x^2 - x + 1 \leq 1$ and $0 < \frac{2x - 1}{2} \leq 1$

$$\Rightarrow x(x - 1) \leq 0 \quad \& \quad 1 < 2x \leq 3$$

$$\Rightarrow x \in [0, 1] \cap x \in \left(\frac{1}{2}, \frac{3}{2} \right]$$

$$\Rightarrow x \in \left(\frac{1}{2}, 1 \right]$$

$$\text{Hence } \alpha + \beta = \frac{1}{2} + 1 = \frac{3}{2}$$

14. Let E_1 and E_2 be two Ellipses with same eccentricity. Where $E_1 \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$) and E_2 is such that it passes through the end points of major axis of E_1 also end points of minor axis of E_1 are foci of E_2 . Find the eccentricity of E_1

(1) $\frac{\sqrt{5} - 3}{2}$

(2) $\frac{\sqrt{5} + 3}{2}$

(3) $\frac{\sqrt{5} + 1}{2}$

(4) $\frac{\sqrt{5} - 1}{2}$

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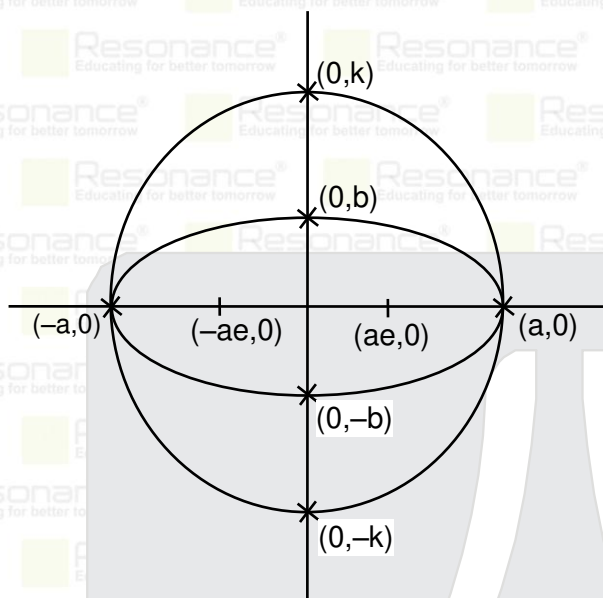
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Ans. (4)

Sol.



Eccentricity of E_1 is $e \Rightarrow e^2 = 1 - \frac{b^2}{a^2}$

Eccentricity of E_2 is $e \Rightarrow e^2 = 1 - \frac{a^2}{k^2}$

So, $e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{a^2}{k^2} \Rightarrow k = \frac{a^2}{b}$ (i)

Also $ke = b$ (ii)

From equation (i) and (ii) $e = \frac{b^2}{a^2}$

Since $e^2 = 1 - \frac{b^2}{a^2} \Rightarrow e^2 = 1 - e$

$\Rightarrow e^2 + e - 1 = 0$

$\Rightarrow e = \frac{\sqrt{5} - 1}{2}$

15. Value of x for which $[e^x]^2 + [e^x + 1] - 3 = 0$ is

- (1) $[e, e^2]$ (2) $[0, e]$ (3) $[e, \ln 2]$ (4) $[0, \ln 2]$

Ans. (4)

Sol. $[e^x]^2 + [e^x + 1] - 3 = 0$

$[e^x]^2 + [e^x] - 2 = 0$

Let $[e^x] = t$

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$$t^2 + t - 2 = 0$$

$$(t+2)(t-1) = 0$$

$$t = 1, -2$$

$$[e^x] : 1, -2 \text{ (-2 is not possible)}$$

$$[e^x] = 1$$

$$x \in [0, \ln 2]$$

16. Evaluates $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$

(1) $e^x \sqrt{\frac{1+x}{1-2x}} + C$

(2) $e^x \sqrt{\frac{1+x}{1-x}} + C$

(3) $e^x \sqrt{\frac{1-x}{1+x}} + C$

(4) $e^x \left(\frac{1+x}{1-x} \right) + C$

Ans. (2)

Sol. $I = \int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$

$$= \int e^x \left\{ \frac{1}{(1-x)^{3/2}(1+x)^{1/2}} + \left(\frac{1+x}{1-x} \right)^{\frac{1}{2}} \right\} dx$$

$$f(x) = \left(\frac{1+x}{1-x} \right)^{\frac{1}{2}} \Rightarrow f'(x) = \frac{1}{(1-x)^{3/2}(1+x)^{1/2}}$$

$$\Rightarrow I = \int e^x (f'(x) + f(x)) dx$$

$$= e^x f(x) + C = e^x \sqrt{\frac{1+x}{1-x}} + C$$

17. Four dice are rolled and the outcomes are put in 2×2 matrices. Find the probability that such a matrix will be non singular and all its entries are different.

(1) $\frac{71}{81}$

(2) $\frac{80}{81}$

(3) $\frac{30}{71}$

(4) $\frac{25}{71}$

Ans. (2)

Sol. $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|X| = ad - bc = 0$$

$$\begin{pmatrix} 1,6 \\ 3,4 \end{pmatrix} \quad \begin{pmatrix} 3,2 \\ 6,2 \end{pmatrix} \left\{ 8 + 8 \text{ possibilities} \right.$$

$$\text{Required probability} = 1 - \frac{16}{6^4} = \frac{80}{81}$$

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18. Find the values of λ & μ for which the system of equations

$$x + y + z = 6$$

$$3x + 5y + 5z = 26$$

$$x + 2y + \lambda z = \mu \text{ has no solution}$$

$$(1) \lambda = 2, \mu \neq 10$$

$$(2) \lambda \neq 2, \mu = 10$$

$$(3) \lambda \neq 3, \mu = 10$$

$$(4) \lambda \neq 2, \mu \neq 10$$

Ans. (1)

Sol. For no solution $\Delta = 0$

$$\Delta = 0$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 3 & 5 & 5 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 1(5\lambda - 10) - 1(3\lambda - 5) + 1(6 - 5) = 0$$

$$\Rightarrow 2\lambda - 4 = 0$$

$$\Rightarrow \lambda = 2$$

$$\Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ 26 & 5 & 5 \\ \mu & 2 & 2 \end{vmatrix} = 0$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 1 \\ 3 & 26 & 5 \\ 1 & \mu & 2 \end{vmatrix} = 1(52 - 5\mu) - 6(6 - 5) + 1(3\mu - 26)$$

$$= 52 - 5\mu - 6 + 3\mu - 26$$

$$\Delta_2 = 20 - 2\mu$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 6 \\ 3 & 5 & 26 \\ 1 & 2 & \mu \end{vmatrix} = 1(5\mu - 52) - 1(3\mu - 26) + 6(6 - 5)$$

$$\Delta_3 = 2\mu - 20$$

Case-I

$$\lambda = 2, \mu = 10 \Rightarrow \Delta = 0, \Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0$$

system of equations are

$$x + y + z = 6$$

$$3x + 5y + 5z = 26$$

$$x + 2y + 2z = 10 \text{ has infinite many solutions}$$

Case - II

$$\lambda = 2, \mu \neq 10 \Rightarrow \Delta = 0, \Delta_1 = 0, \Delta_2 \neq 0, \Delta_3 \neq 0$$

system has not solution

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19. The number of elements in the set $\{x \in \mathbb{R} : (|x| - 3)|x - 4| = 6\}$ is equal to

- (1) 3 (2) 4 (3) 2 (4) 1

Ans. (3)

Sol. $(|x| - 3)|x - 4| = 6$

Case - 1

$$x \geq 4$$

$$(x - 3)(x - 4) = 6$$

$$x^2 - 7x + 6 = 0$$

$$(x - 1)(x - 6) = 0$$

$$x = 1, 6 \Rightarrow x = 6$$

Case - 2

$$0 < x < 4$$

$$(x - 3)(4 - x) = 6$$

$$x^2 - 7x + 18 = 0$$

$D < 0$, No solution

Case - 3

$$x \leq 0$$

$$(-x - 3)(4 - x) = 6$$

$$(x + 3)(x - 4) = 6$$

$$x^2 - x - 18 = 0$$

$$x = \frac{1 \pm \sqrt{73}}{2}$$

$$x = \frac{1 - \sqrt{73}}{2}$$

20. If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi} - \left[\frac{x}{\pi}\right]\right)}} dx = \frac{\alpha\pi^3}{1 + 4\pi^2}$, $\alpha \in \mathbb{R}$ where $[x]$ is greatest integer function, then α is

- (1) $50(e - 1)$ (2) $150(e^{-1} - 1)$ (3) $200(1 - e^{-1})$ (4) $100(1 - e)$

Ans. (3)

Sol. $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}} dx$

$$\Rightarrow 100 \int_0^{\pi} \frac{\sin^2 x}{e^{x/\pi}} dx$$

$$\Rightarrow 50 \int_0^{\pi} e^{-x/\pi} [1 - \cos 2x] dx$$

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